

Preface

This is a book about applied geophysics written for two different audiences. It is meant for university students at the intermediate level in mathematics and physics who are preparing for future professional work in geophysics; it is also intended for persons working as professionals in geophysics or in related sciences who wish to brush up on the fundamentals of their specialty or broaden their acquaintance with others. The aim is to help build the mathematical and physical foundations for interpreting observational data in the areas of applied seismology, gravimetric and magnetometric surveying, and electrical (including electromagnetic) exploration. Wherever possible, we have tried to keep the two audiences impartially in view; but where their interests diverge—as inevitably they must—our attention is directed toward the student group rather more than to the professionals. However, the academic tone of this book should not obscure its essential function which is to try to relate theory to practice. Indeed if it were not so, the title would be difficult to justify.

Our purpose in writing this book has been to fill a gap in the literature of applied geophysics which has been in existence for many years and which today is widening inexorably. This is the gap in technical level between the introductory texts (written mostly for nonspecialists) and current research journals. There is not at the present time a single up-to-date book in the English language which even partially bridges it. In teaching applied geophysics to university students, this situation has for many years been an embarrassment to us and a

Interpretation Theory in Applied Geophysics

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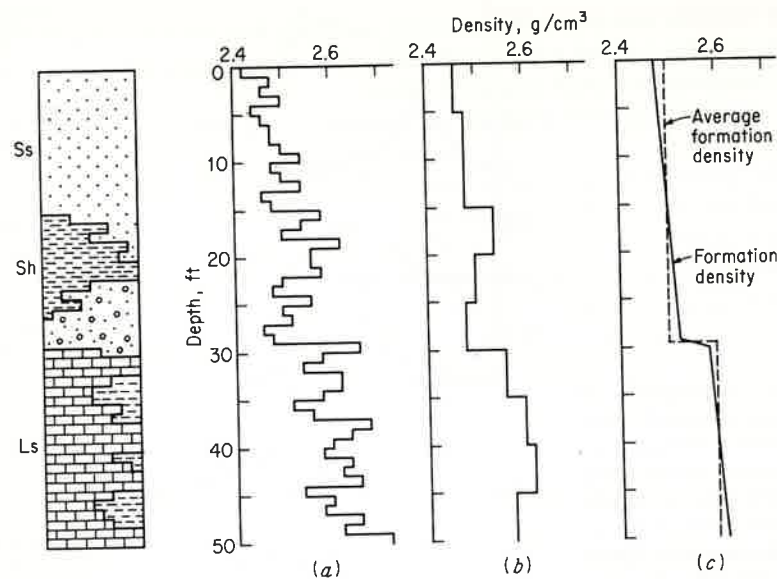


Fig. 7-4 Formation and average densities within a stratigraphic sequence.

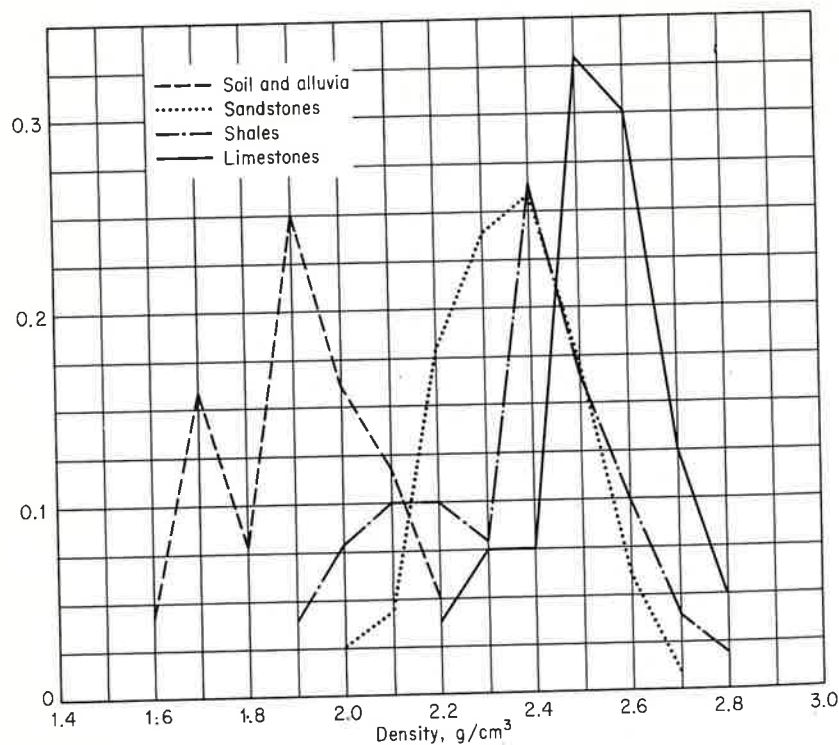


Fig. 7-5 Histograms of small-specimen bulk densities of various kinds of sedimentary rocks. [From Birch (4).]

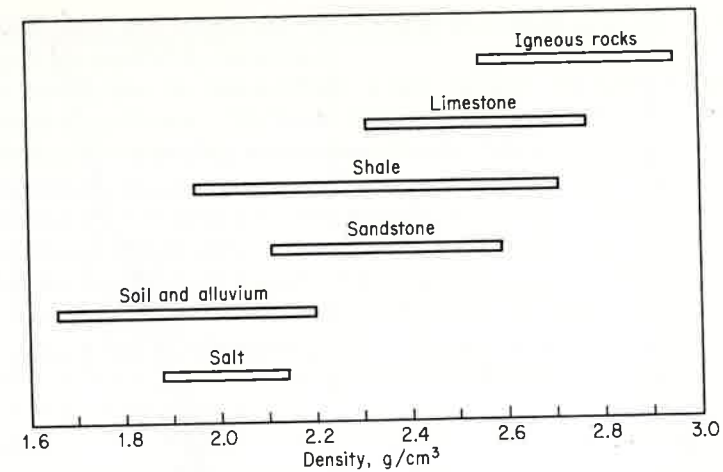


Fig. 7-6 80 percent fiducial limits of small-specimen bulk densities of various kinds of rocks. [From Birch (4).]

groupings and are plotted without regard to age or depth of burial, and because sampling statistics were not provided with the tables, each entry was accorded the same weight. Two things are clear: there are extensive overlaps among all groups, and in most cases the frequency distributions are noticeably skewed. The overlapping is best demonstrated by the 80 percent fiducial limits derived from these data and shown in Fig. 7-6. The skewness probably can be explained for shales by the nonlinearity of the porosity-depth relation, and for limestones, by the sampling of reefal materials having anomalously high porosity. In any event, even if the lithological aspects of two contiguous formations are known, it is evidently impossible to predict with confidence the difference in average density between them, without having access to other geological information.

Should seismic information be available within the area, however, then the situation is less bleak. As we have already indicated that both seismic-wave velocities and bulk densities are affected by lithification accompanying age and compaction, it should not be surprising to discover a tendency for these two physical properties to increase together. Figure 7-7 shows a plot of seismic velocity measurements versus bulk density determinations which shows that they do. This graph, prepared by Nafe and Drake (5), contains data from a large number of core samples taken throughout a considerable range of depths and from widely distributed points. The mean curve, fitted by least squares, can be used for predicting the bulk density of a formation from its seismic velocity when only the latter is known. Although there is a considerable scatter about this curve, it is a useful device for gravity interpretations when no other information about the densities is available.

Table 12-2 Susceptibility of Common Rocks*

Rock type	Number of samples	Percent having volume susceptibility, emu			
		$k < 10^{-4}$	$10^{-4} < k < 10^{-3}$	$10^{-3} < k < 4 \times 10^{-3}$	$k > 4 \times 10^{-3}$
Basic effusive	97	5	29	47	19
Basic plutonic	53	24	27	28	21
Granites and allied rocks	74	60	23	16	1
Gneisses, schists, slates	45	71	22	7	0
Sedimentary rocks	48	73	19	4	4

* From L. B. Slichter (6).

direction parallel to it. Variations of several percent are not uncommon, and extreme cases arise in finely banded magnetite or pyrrhotite. There the demagnetization factor is for the most part 4π in the direction transverse to the structure and zero in a direction parallel to it. Thus the transverse susceptibility will be just $p/4\pi$, while the parallel susceptibility will be limited only by the defect structures in the ferromagnetic grains and by the appropriate demagnetizing factor of the outer envelope of the mineralized region.

Several attempts have been made to formulate statistical "laws" relating the bulk susceptibility of rocks to petrological parameters. A useful guide, taken from "Handbook of Physical Constants" (6), is given in Table 12-2. It clearly demonstrates the tendency of magnetite to concentrate in mafic rocks, which therefore exhibit a higher susceptibility than the more silicic types.

Mooney and Bleifuss (7) have measured the susceptibilities of a suite of Precambrian rocks from Minnesota, and the averages they obtained for different rock types are shown in Table 12-3.

Table 12-3

Rock type	No. of samples	Mean k , emu
Basalt	37	2.95×10^{-3}
Diabase	19	2.59×10^{-3}
Rhyolite	5	1.12×10^{-3}
Gabbro	37	0.99×10^{-3}
Granite	31	0.47×10^{-3}
Other acid intrusives	17	0.35×10^{-3}
Ely greenstone	15	0.09×10^{-3}
Slates	26	0.05×10^{-3}

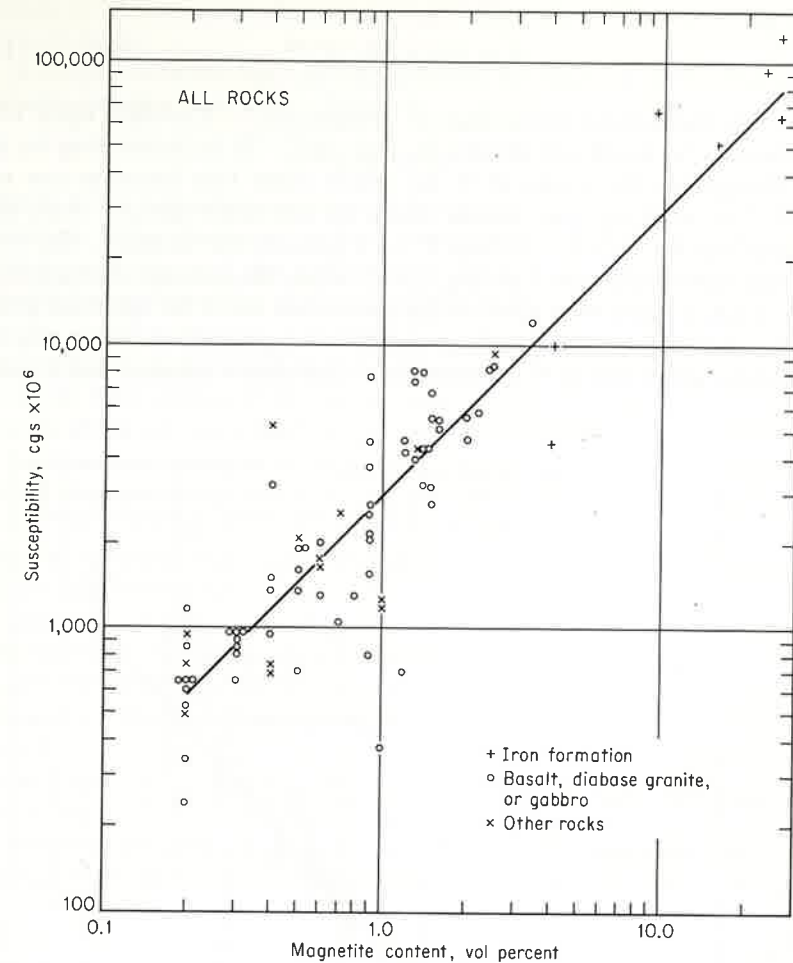


Fig. 12-6 Data from which the empirical formula for susceptibility $k = 2.89 \times 10^{-3} V^{1.01}$ was derived. [Mooney and Bleifuss (7).]

They also have derived a formula relating the bulk susceptibility to the volume of magnetite found by crushing, magnetic separation, and chemical analysis for iron. It is

$$k = 2.89 \times 10^{-3} V^{1.01} \quad (12-3)$$

where V is the volume percentage of magnetite, and the data on which it is based are shown in Fig. 12-6.

Balsley and Buddington (8) have related the susceptibility of a suite of Adirondack rocks to the fractional volume of all the minerals visually identified as "magnetite," which would generally include any Fe-Ti oxide minerals of spinel structure. Their empirical formula for the bulk suscepti-

Table 12-4

Area of origin	Great soil group	k, emu × 10 ⁻⁶		Mag-netic frac-tion, ppm	Composition of the most magnetic fraction	
		Orig-i-nal	Re-sidual			
C. Texas	Reddish chestnut	25	20	33	.35 magnetite	.6 quartz
S. Texas	Rendzina	10	7	17	.45 magnetite	.55 quartz
Texas coast	Sand	5	2	18	.5 magnetite	.5 quartz
Texas coast	Semi-bog	5	5	?	.25 magnetite	.1 limonite
					.5 quartz	.15 gypsum
E. Texas	Yellow podzolic	10	3	?	.5 magnetite	.4 quartz
					.09 ilmenite	.01 gypsum
E. Texas	Yellow podzolic	15	2	215	.75 magnetite	.25 quartz
C. Texas	Rendzina	350	330	1050	.1 magnetite	.9 quartz
W. Texas	Reddish brown	180	35	75	.8 magnetite	.15 ilmenite (?)
					.05 quartz and gypsum	
N. New Mex.	Sand	80	5	700	.95 magnetite	.05 quartz
C. Colo.	Brown	1080	200	2400	.7 magnetite	.3 quartz

soil susceptibilities. The majority of the large magnetic particles (i.e., those visible under the microscope) were extracted from each soil specimen and identified. Nearly all particles turned out to be magnetite. The susceptibility of the residuum was then measured to see what fraction of the original susceptibility had been contributed by the extract. In four of the ten cases it was found to have produced less than half, and thus the susceptibility of these soils is probably due to ultra-fine-grained maghemite, as suggested by Le Borgne. In the other six cases the magnetite content was more important.

Cook and Carts made no direct measurements of the NRM of the soils they sampled, but their work did indicate that it may be appreciable. Le Borgne has found the soils containing ultra-fine-grained magnetic material to be very susceptible to VRM, indicating that the particles are mono-domain grains on the verge of superparamagnetism. Such material, unless disturbed regularly by operations such as cultivation, will almost certainly exhibit an appreciable magnetic remanence.

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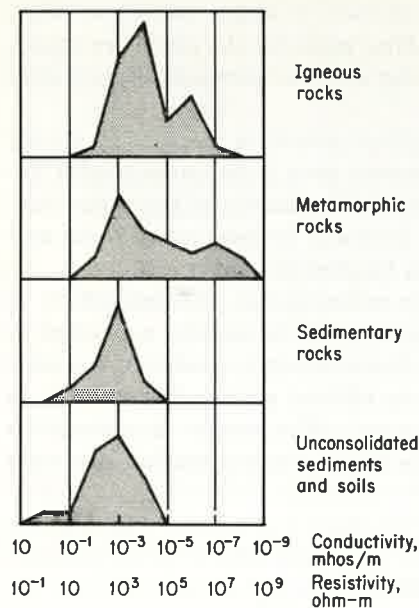


Fig. 13-4 Histograms of resistivity measurements listed in "Handbook of Physical Constants" (11).

best easily available tables of measurements are given by Slichter and Telkes in the "Handbook of Physical Constants" (11). As a summary of these tables, histograms of the conductivities of igneous, metamorphic, and sedimentary rocks and of unconsolidated sediment and soils are shown in Fig. 13-4. These diagrams have no very simple statistical meaning, since the observations do not form any sort of representative or random sample. There is also little guarantee that the natural moisture was retained or restored to each sample before measurement, so that some measurements may show anomalously low conductivity values. Yet in spite of this, the general observations that we mentioned earlier are confirmed. The sedimentary rocks and unconsolidated materials are distinctly more conductive than most of the igneous rocks on account of their higher porosities and moisture content. Metamorphic rocks are more variable because they have a very wide range of porosities, extending from values similar to those of sediments to virtually zero among the crystalline materials that have been reformed under very high containing pressures. The soils appear to have conductivities similar to those of sedimentary rocks, in spite of their greater porosities. The explanation of this phenomenon is very likely that these materials are saturated by fresher, purer water than the older and more deeply buried sediments. This may also account for the fact that logging measurements in oil wells generally reveal resistivities in the range 1 to 1,000 ohm-m, which is decidedly lower than the great majority of the sample measurements. The general increase of temperature with depth may also

be a factor since the conductivity of electrolytes is very temperature sensitive.

Because of the very widespread use of electrical logging methods in oil wells, a considerable amount of research has been done toward establishing a relation between the resistivity and the porosity, saturation, and other observable attributes of certain types of sedimentary rock. Those that have been the most extensively studied are the sandstones and carbonate rocks, which commonly form petroleum reservoirs. For clean rocks (i.e., those without clay in their pores) it has been found that a good estimate of the resistivity in situ is given by the following relationship:

$$\text{Resistivity } \rho = IF\rho_w$$

where F = the formation factor

I = the resistivity index

ρ_w = the resistivity of the water in the formation

In terms of porosity and saturation, the formation factor and the resistivity index are found to have the following forms:

$$F = aP^{-m} \quad \text{and} \quad I = S^{-n}$$

where P = the porosity of the rock formation

S = fractional amount of saturation of pore spaces

a, m, n = empirical constants

For clean sandstones, values between 0.6 and 1.0 for a , of 2.0 for m , and of 2.0 for n are generally accepted; and for carbonate rocks $a = 1.0$, $m = 1.8 - 2.6$, and $n = 1.5 - 2.2$, depending upon the type. Those which have large pores and cavities generally exhibit the larger values of m , while the larger values of n seem to be an attribute of the more dense and compact varieties.

The presence of clay in the pores of a rock has a considerable effect upon its conductivity. The clay minerals and other hydrous substances such as serpentine are generally found to be rather good conductors. In sandstones a small quantity of clay tends to make the rock much more conductive at low water saturations, and often in such circumstances a value of about 1.5 rather than 2 is indicated for the exponent m in the resistivity formula. Although a dry clay mineral is not itself an unusually good conductor, a small amount of excess water can make it so through the action of ion exchange.

Shales are generally worse conductors than the more porous rocks, and they are also often noticeably anisotropic. Minute mica grains and other platy minerals tend to lie in parallel bedding planes, giving higher conductivities along these planes than across them. Ratios of parallel to transverse conductivity as high as four are not uncommon.

The determining factor in the electrical properties of all these materials is the conductivity of the electrolyte which permeates the rock. In dilute

Table 13-1

Mineral	Conductivity σ , mhos per meter					
	Pure crystalline form		Natural mineralization			
	In direc- tion giving maximum	In direc- tion giving minimum	Approx. 100 %		Approx. 50 %	
			Mean	s	Mean	s
Pyrite FeS ₂			10 ²	1.2	(% mineralization uncertain)	
Pyrrhotite FeS(Fe ₇ S ₈)			3 × 10 ⁴	0.3	6 × 10 ³	
Chalcopyrite CuFeS ₂			2 × 10 ³	0.4		
Arsenopyrite FeAsS			3 × 10 ³	0.3		
Galena PbS	5 × 10 ⁴		10		5	
Magnetite Fe ₃ O ₄	10 ⁴		3	1	(% mineralization uncertain)	
Graphite C	10 ⁸	10 ²	5 × 10 ³		100	1.0
					on shales containing only a few % C	
Hematite Fe ₂ O ₃		insulator	0.1	0.8		

Mean = geometric mean = $\exp_{10} \frac{1}{n} \sum \log_{10} \sigma_i$.

s = standard deviation of logarithm of conductivities (rough estimate).

constant, the variations covering only about one order of magnitude in the vicinity of 10⁴ mhos/m. This is confirmed by experience in the field, which indicates that bodies which contain any substantial amount of pyrrhotite are nearly always extremely good conductors.

Chalcopyrite is a mineral that seems to be comparable to pyrrhotite, but it has a mean conductivity of only 2 × 10³ mhos/m. The scatter of values is intermediate between pyrite and pyrrhotite, covering about two decades. The same remarks apply also to *arsenopyrite*.

Galena in small, pure specimens is apparently an excellent conductor, having a conductivity of 10⁴ mhos/m. However, its cubic habit seems to make it a poor conductor in polycrystalline specimens even in fairly concentrated ores, since the grains are apparently unlikely to be interconnected. It appears that *galena* mineralization by itself is a poor conductor, but when mixed with other metallic minerals which tend to connect the grains, it may possibly conduct quite well.

Magnetite seems to be somewhat similar to *galena* in that its crystal conductivity is high, but in polycrystalline or even massive form it is not a good conductor. Again this is probably due to the tendency to form in discrete, euhedral crystals.

Graphite in pure crystalline form is an excellent conductor, having a conductivity of 10⁶ mhos/m in the basal plane and 10² mhos/m across it. Polycrystalline graphite appears to be intermediate between these two values at 5 × 10⁴ mhos/m. Its most remarkable feature is the amazing way in which it remains connected even when present in amounts of only a few percent. While 2 percent pyrite will have virtually no effect upon the conductivity of a rock, 2 percent graphite will give it a very noticeable conductivity increase.

Hematite is not normally a conductor. However, it appears that the presence of impurities can induce an appreciable conductivity at least in the specular form of this mineral—up to the range 1 to 10² mhos/m. Because of the large size of some hematite bodies this may have important consequences in electrical surveys.

Zincblende is also an insulator. This is certainly true of pure ZnS and probably also applies to the varieties containing iron (*sphalerite*). It is possible (although this has not actually been demonstrated) that impurities such as an excess of iron or other metals might make it a conductor.

Other minerals which have been found to be appreciably conductive but which have not been so well studied are *bornite* (CuFeS₄), *chalcocite* (Cu₂S), and *covellite* (CuS), which seem to have conductivities comparable to that of chalcopyrite, *ilmenite* (FeTiO₃), *molybdenite* (MoS₂), and the manganese minerals *hollandite* and *pyrolusite*, which seem to have conductivities that are at least as great as 1 mho/m. In addition, it is hardly necessary to mention that the native metals are excellent conductors, the conductivities approaching those of their commercial forms. Metallic silver has a conductivity of about 6 × 10⁷ mhos/m; copper, 5.8 × 10⁷ mhos/m; and iron, approximately 5 × 10⁶ mhos/m.

The third class of conductors we shall call the *structural* conductors. These have little to do with rock type, porosity, or metallic mineral content but occur as a result of certain geological structures superimposed on the rocks. Chiefly, these structures are of the linear type, such as faults, shear zones, or contact fracture zones, where fracturing has increased the space available to ground water. Features such as these are of considerable importance to electrical surveys because they may be very well connected over great areas—a property which often compensates for their moderate bulk conductivity. In addition, they are often found to be the locus of alteration and weathering which has produced clay and other hydrous minerals. These minerals in turn can increase the effective conductivity of the water in the zones.

Such structures are often difficult to observe, except perhaps in mines.